Examples: State the derivatives of each function

a) $f(x)=x^{100}$

$f'(x)=100x^{99}$

b) $f(x)=\pi$

$f'(x)=0$

c) $f(x)=-7.39$

$f'(x)=0$

d) $y=9x^5$

$y'=45x^4$

e) $f(x)=(2x)^4$

$f(x)=16x^4$

$f'(x)=64x^3$

f) $y=(x+1)^2$

$y=x^2+2x+1$

$y'=2x+2$

g) $f(x)=x^2(x^2+3)$

$f(x)=x^4+3x^2$

$f'(x)=4x^3+6x$

Examples: Differentiate

a) $f(x)=(1-2x)^3$

$f(x)=-8x^3+12x^2-6x+1$

$f'(x)=-24x^2+24x-6$

b) $f(x)=(x+2)(x^2-1)$

$f(x)=x^3+2x^2-x-2$

$f'(x)=3x^2+4x-1$

c) $s=\dfrac{3t^5-2t^2+5t^{-1}}{t^2}$

$s=3t^3-2+5t^{-3}$

$s'=9t^2-15t^{-4}$

$s=9t^2-\dfrac{15}{t^4}$

d) $u=\sqrt{\dfrac{2}{x}}+\sqrt{\dfrac{x}{3}}$

$u=\sqrt{2}\,x^{-\frac{1}{2}}+\dfrac{1}{\sqrt{3}}x^{\frac{1}{2}}$

$u'=-\dfrac{\sqrt{2}}{2}x^{-\frac{3}{2}}+\dfrac{1}{2\sqrt{3}}x^{-\frac{1}{2}}$

$u'=-\dfrac{\sqrt{2}}{2\sqrt{x^3}}+\dfrac{1}{2\sqrt{3x}}$

1. Differentiate the following functions

a) $f(x) = 3\sqrt{x} - \frac{1}{\sqrt{x}}$

$f(x) = 3x^{\frac{1}{2}} - x^{\frac{-1}{2}}$

$f'(x) = \frac{3}{2\sqrt{x}} + \frac{1}{2\sqrt{x^3}}$

b) $y = \sqrt{x}\left(x^3 - x^{\frac{1}{2}} + 5\right)$

$y = x^{\frac{7}{2}} - x + 5x^{\frac{1}{2}}$

$y' = \frac{7}{2}x^{\frac{5}{2}} - 1 + \frac{5}{2}x^{\frac{-1}{2}}$

$y' =$ [redacted]

$y' = \frac{7\sqrt{x^5}}{2} - 1 + \frac{5}{2\sqrt{x}}$

c) $f(x) = \frac{2x^5+5x^2+x}{3x^2}$

$f(x) = \frac{2x^3}{3} + \frac{5}{3} + \frac{x^{-1}}{3}$

$f'(x) = 2x^2 - \frac{1}{3x^2}$

2. Find an equation of the tangent line to the curve $y = x^4 + 2x^2 - x$ at point $(1, 2)$.

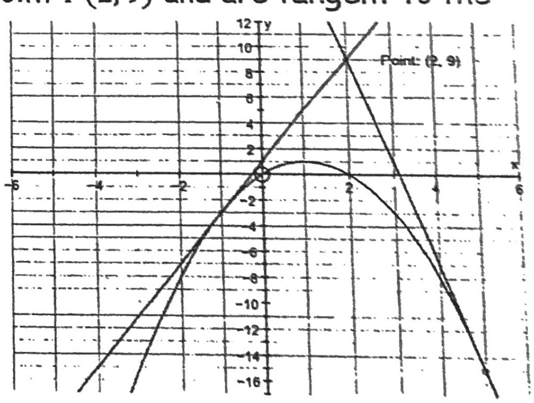
$\frac{dy}{dx} = 4x^3 + 4x - 1$

$\left.\frac{dy}{dx}\right|_{x=1} = 7$

$\therefore\ y - y_1 = m(x - x_1)$

$y - 2 = 7(x - 1)$

3. Find the equations of both lines that pass through the point $P(2, 9)$ and are tangent to the parabola $y = 2x - x^2$. $[y = 4x + 1, y = -8x + 25]$



$y' = -2x + 2$

note:

$P(2, 9)$ is not on the curve.

$y - y_1 = m(x - x_1)$

$2x - x^2 - 9 = (-2x + 2)(x - 2)$

$-x^2 + 2x - 9 = -2x^2 + 6x - 4$

$x^2 - 4x - 5 = 0$

$(x - 5)(x + 1) = 0$

$\therefore\ x = 5, -1$

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① $y - 9 = -8(x - 2)$

$y = -8x + 25$

② $y - 9 = 4(x - 2)$

$y = 4x + 1$

Ex. 1: Differentiate: $h(x)=\sqrt{x}(1+x)$

$$h(x) = x^{\frac{1}{2}} + x^{\frac{3}{2}}$$

$$h'(x) = \frac{1}{2}x^{-\frac{1}{2}} + \frac{3}{2}x^{\frac{1}{2}}$$

$$h'(x) = \frac{1}{2\sqrt{x}} + \frac{3\sqrt{x}}{2}$$

Ex. 2: Find the equation of the tangent line to the curve

$f(x)=(\sqrt{x}+2)(1-2\sqrt{x}+3x)$ at $x=1$.

$$f'(x) = \frac{1-2\sqrt{x}+3x}{2\sqrt{x}} + (\sqrt{x}+2)\left(-\frac{1}{\sqrt{x}}+3\right)$$

$$f'(1) = \frac{1-2+3}{2} + (1+2)(-1+3)$$

$$f'(1) = 1 + 6$$

$$f'(1) = 7$$

$$y - y_1 = m(x - x_1)$$

$$\therefore\ y - 6 = 7(x-1)$$

Ex. 1: Differentiate $h(x) = \frac{x^2+2x-3}{x^3+1}$.

$$h'(x) = \frac{(2x+2)(x^3+1) - (x^2+2x-3)(3x^2)}{(x^3+1)^2}$$

$$h'(x) = \frac{2x^4 + 2x + 2x^3 + 2 - (3x^4 + 6x^3 - 9x^2)}{(x^3+1)^2}$$

$$h'(x) = \frac{-x^4 - 4x^3 + 9x^2 + 2x + 2}{(x^3+1)^2}$$

Ex. 2: Find $\frac{dy}{dx}$ if $y = \frac{\sqrt{x}}{1+2x}$.

$$y' = \frac{\frac{1}{2\sqrt{x}} \cdot (1+2x) - \sqrt{x} \cdot (2)}{(1+2x)^2}$$

$$y' = \frac{1 + 2x - 4x}{2\sqrt{x}\,(1+2x)^2}$$

$$y' = \frac{-2x+1}{2\sqrt{x}\,(1+2x)^2}$$